
Erik Anderson

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Dear Hiring Team,

I'm writing to express my interest in the Software Application Developer position within the Robotics Department at the University of Michigan. I recently graduated from the University with a degree in Computer Science and Italian Language and Literature, and I'm drawn to this role because it combines my experience in reliable software development with my interest in supporting research and education.

In my work with the University's Information and Technology Services, I've supported and maintained over 300 campus computers and handled a high volume of support requests through Amazon Connect. I learned how to stay composed under pressure, communicate clearly with non-technical users, and document solutions that make future troubleshooting faster and more consistent. This work shaped my ability to bridge technical systems with real user needs.

On the development side, I've built and tested web applications using Python, JavaScript, and SQL. My projects include a distributed key-value store built with Paxos consensus for fault tolerance and a full-stack web app that ingested and processed large datasets using Flask and PostgreSQL. These experiences taught me to write clear, maintainable code, test thoroughly, and always think about how to make tools intuitive for the people who use them.

What draws me most to this position is the collaborative environment and the emphasis on continuous improvement within Michigan Engineering. I enjoy solving practical problems for others, whether it's debugging code, designing a workflow, or improving a support process. I'm comfortable working independently but also enjoy learning from others and contributing to a team that values thoughtful engineering and clear communication.

Thank you for your time and consideration. I would be grateful for the opportunity to contribute to the Robotics Department's work and support the innovative projects happening here at Michigan.

Sincerely,

Erik Anderson

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EDUCATION

University of Michigan

Aug 2020 - Dec 2024

B.S., Computer Science & Italian Language and Literature

- **GPA:** 3.4/4.0
- **Coursework:** Web Systems, Information Retrieval & Web Search, Conversational AI, Distributed Systems, User Interface Development, Discrete Mathematics, Data Structures & Algorithms, Statistics & Data Analysis

University of Bologna

Jan 2023 - Jun 2023

Advanced Italian Language Program

WORK EXPERIENCE

InvolveMINT

Jul 2025 - Present

Software Engineer Intern

- Developed and tested full-stack TypeScript/Node.js features for a public-good microtransactions platform, enhancing backend validation, API integrations, and user workflow reliability.
- Optimized PostgreSQL-based credit ledger and voucher systems by debugging and refining transaction processes, contributing to improved system accuracy.

University of Michigan Information and Technology Services

Dec 2023 - Present

Tech Consultant

- Diagnosed and resolved technical issues for over 300 campus lab computers, ensuring reliable application support and adhering to established support protocols.
- Provided consultative support and troubleshooting at the campus tech shop, effectively gathering requirements from users and documenting hardware, software, and network solutions.
- Managed and addressed a high volume of support tickets via Amazon Connect, employing troubleshooting techniques and clear communication to meet service benchmarks.

RELEVANT EXPERIENCE

Make n' Model

Jan 2024 - Apr 2024

- Built a personalized model kit recommendation system using Flask and SQL, with Naive Bayes-based difficulty scoring (F1: 0.48) and real-time inventory filtering.
- Integrated Google Cloud OCR and FastText to streamline multilingual PDF parsing, cutting document processing time from 15 to under 5 minutes across over 100GB of instruction manuals.

Distributed Systems and Fault-Tolerant Services (PAXOS)

Sep 2024 - Dec 2024

- Implemented a sharded, fault-tolerant key/value store with Paxos consensus, quorum agreement, and dynamic reconfigurations.
- Designed garbage collection and safe state replication across distributed nodes under real-world fault conditions

Insta485

Sep 2023 - Nov 2023

- Developed an Instagram-style web app with infinite scroll, likes/comments, session auth, and Flask-backed routing; deployed to AWS EC2 for live access and testing.
- Engineered async state syncing, modular API handling, and lifecycle-safe hooks for responsive state updates and secure user session handling.

Google-Style Search Engine

Nov 2023 - Dec 2023

- Implemented a scalable tf-idf search engine with a segmented inverted index, index server, and REST API, mirroring Google-style architecture.
- Built a custom MapReduce framework to execute distributed processing, leveraging OS-level concurrency, fault tolerance, and network coordination.

WaldoGPT

Sep 2024 - Nov 2024

- Developed WaldoGPT, an AI-powered tool to streamline video content analysis by integrating automatic speech recognition and object identification, enabling users to transcribe, index, and search video content efficiently using OpenAI's Whisper API and other AI models.

TECHNICAL SKILLS

- **Languages:** Python, Go, JavaScript, TypeScript, C, C++, HTML, CSS, SQL, Bash
- **Frameworks & Tools:** React, Flask, Axios, Express, Node.js, Docker, PostgreSQL, MongoDB, Git, GitHub Actions (CI/CD), Amazon Connect, AWS EC2, Figma
- **Systems & Concepts:** Web Architecture, Distributed Systems, System Design, Scalability, Fault Tolerance, Authentication, Sharding, Load Balancing, SDLC, Agile Development, Sockets, REST APIs, WebSockets, Linux, Web Application Development